

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Name	Smart Lender

1. PROJECT OVERVIEW	
Section	Description
Objective	To develop an AI-powered Smart Lender system that predicts loan eligibility instantly based on applicant details, helping both customers and financial institutions make faster, accurate, and data-driven lending decisions.
Scope	The project delivers a web application that enables users to input personal, financial and employment details and receive an instant loan eligibility prediction along with eligibility score, approval probability and recommended loan amount range.
Target Users	Individual loan applicants (salaried/self-employed), banks, NBFCs and loan officers who need quick and reliable loan eligibility assessment.
Goal	Reduce manual verification workload, minimize rejection rates and turnaround time, and provide a transparent, accurate and user-friendly loan eligibility prediction system to improve customer experience and operational efficiency.

2. PROBLEM STATEMENT	
Section	Description
Description	Loan approval process in traditional banking and NBFC systems is slow, complex and highly manual. Applicants must submit multiple documents, undergo lengthy verification and still face uncertainty about eligibility before applying.
Impact	High rejection rates, long turnaround time and lack of transparency lead to poor customer experience. Financial institutions also face high operational costs and increased risk of defaults due to inefficient assessment.
Key Problem	- No instant way for applicants to know their loan eligibility. - Manual verification causes delays and human errors. - Multiple factors affecting eligibility are difficult to analyze manually. - Lack of data-driven insights results in higher rejection rates and dissatisfied customers.

3. PROPOSED SOLUTION

Section	Description
Approach	Develop an AI/ML based loan eligibility prediction model using historical loan data. The model analyzes applicant details and predicts eligibility, approval probability and suitable loan amount. The prediction engine is integrated into a user-friendly web application for real-time results.
Key Features	- Machine Learning model for accurate loan eligibility prediction. - User-friendly interface for input and results. - Instant eligibility status with approval probability and loan amount range. - Data visualization and insights for better decision-making. - Secure storage of applications and prediction history. - Admin dashboard for managing data and monitoring system performance.
Expected Outcome	- Faster loan eligibility assessment. - Improved accuracy and reduced manual effort. - Higher customer satisfaction and transparency. - Better decision-making for financial institutions.

4. RESOURCE REQUIREMENTS

Resource Type	Description	Specification / Allocation
Hardware	System for development and testing	Processor: Intel Core i3 or above, RAM: Minimum 4 GB, Storage: Minimum 10 GB free space
Internet	Internet Connectivity	Required for downloading packages, datasets and deployment
Operating System	Platform for development	Windows / Linux / macOS
Framework	Web framework	Flask (Python)
Libraries	Core libraries for ML & data handling	scikit-learn, pandas, numpy, SQLAlchemy, joblib
Development Environment	IDE / Environment	Visual Studio Code, Jupyter Notebook / PyCharm
Data Source, Size, Format	Dataset for model training	Historical loan data (bank/NBFC dataset), CSV format